

Taffazzel Hossain

Big-Data Engineer

Mob:07388331609 | Email – tafaz_hossain@yahoo.com

LinkedIn- <https://www.linkedin.com/in/taffazzel-hossain/>

Computer engineer with around eight years of hands-on industry experience in Data Engineering with focus on solving problems related to banking and finance. Having delivered many data engineering projects in cross-functional teams as client-facing consultant and in team as mid-level lead having mentored juniors. Track record of successful delivery of big data pipelines, platforms, migration and ETL projects for clients in the UK and Europe including large banks. Apart from achieving the global certifications in Data Engineering, a data structure & algorithm and leetcode enthusiast.

Certifications:



Tools and Technologies:

- Development - **Python(3.9)** Test Driven Development, **PySpark**, Scala Spark, **SQL**
- AI Code Assistant- **Claude, Co-Pilot**
- Big Data- **Azure databricks**, Data lake,Hadoop,Spark(**PySpark 3.5.1**), **SparkSQL**, Airflow, Docker, Kubernetes, YARN, Hive.
- Big Data file formats: **Parquet,Delta**,Avro,ORC.
- AWS: ElasticSearch, S3, EC2, EMR, RDS, Athena, Glue (ETL/Catalog), DMS, Lambda, ECS, VPC etc.
- Business Intelligence: Power BI, AWS Quicksight
- Databases –MySQL
- Automation– CloudFormation, Terraform
- Version Control: Git (GitHub, Gitlab, Bitbucket).
- Others: JIRA, Confluence, CI/CD-Azure Devops, Shell Script, Machine Learning (basics)

Experience:

Data Engineer at Synechron Consulting

July 2022-Present

Projects involved:

HSBC, London

Observability, Big Data Engineer

October,2025-Present

- Development:

- On Hadoop Cloudera platform, having utilized Splunk data, built several scalable in-production big-data pipelines dealing with large-scale datasets.
- Creating Data Lake on HDFS. Building platform for analysis using Hive backend being Spark/Tez/Map-reduce engine and performing benchmarking and tuning for different use cases.
- Rule based custom ingestion pipeline and cataloguing and observability dashboard.
- Live Stream: Dealing with live stream dataset using Kafka and spark
- Test Driven Development: TDD approach with software development best practices.

Technologies/Platforms Used: Splunk, Hadoop (Cloudera), Hive, Kafka, Spark, Python, Github, Airflow, JupyterHub.

Citi Bank, London

Optima DX, Python Data Engineer

April,2025 – October,2025

- **Development:** Built python application aiming to deal with large dataset on a very specific use case using NumPy, Pandas, Polars.
- **Performance Tuning:** Benchmarking and Performance tuning existing application codes to make the application performant.
- **Practices:** Used Test Driven Development (TDD) in Python with github versioning and utilized CI/CD pipelines for deployments at various stages- Dev/UAT/Prod. All work is done under Agile process.

Technologies Used: Python 3.9, SQL, Unix, Yaml, Github, Bitbucket

Spark developer at Synechron/HSBC, London

Feb 2024 - Feb 2025

Project:

ESG Data-Governance team in Finance lead disclosure (FLD):

- **Data Quality and Validation:** Implemented automated Data Quality Frameworks (completeness, accuracy, consistency etc) for ESG Data sources such as emission, Bio-diversity supply chain etc from various clients/vendors which help detect data anomalies and ensures aligned with regulation and end goal was to help and support serve the loan best possible to the global and wider environmental commitments.
- **Metadata Management:** Maintained and enriched metadata for ESG datasets (Data lineage, business definitions, data owners etc)
- **Data Lineage:** Developed end-to-end data lineage from ingestion -> transformation -> reporting, enabling transparency in ESG metric calculations.
- **Master & Reference Data Management:** Created and maintained centralized reference datasets. Maintained data consistency across departments.
- **Data Security and Access Control: Implemented** role-based access and secure data zones (raw/curated/production) in Data Lake.

Technical deliveries:

- **Spark Development:** In Spark, built production ready ETL pipelines (data assets) from scratch to automate the data ingestion, transformation applying complex logic on financial

dataset and load to and from different data platforms for various ESG clients on different use cases.

- **Practices:** Worked in agile environment. Implemented OOPs, maintained Test Driven Development (TDD) with GitHub best practices. Maximized reusability.CI/CD through Jenkins and scheduling and orchestration through Airflow. Monitored and debugged ETL data pipelines through Hadoop yarn logs.
- **Problem Solving:** Worked on problem solving for various data assets using PySpark and Python translating from business analysts produced functional specification document to running Spark workflow showcasing result to business users. Successfully implemented complex logics in Python/PySpark and then replicated them to Scalable Engineering pipeline.
- **Teamwork:** Coordinated with data analysts, business analysts, delivery lead along with other data engineers including various leadership.
- **Mentoring responsibility:** Team members' code review and approve. Mentored junior team members.
- **Performance Tuning:** Worked on performance tuning. Yarn cluster setup tuning and Spark code rewritten to get optimum results on efficiency.
- **Other contribution:** Internally helped Synechron leadership to build a structure of interview process for the new recruits. Created a **question and answer** set in the overlapping area of Big Data Engineer/ Big Data Developer/ Spark Developer/AWS Data Engineer. Currently **being used** by the recruiters internally.

Technologies Used: Spark, Python, SQL, Jupiter notebook, Hadoop cluster- Cloudera, Jenkins, Airflow, HDFS, Hive warehousing, GitHub, Parquet.

Big data developer at Synechron/HSBC, London:

July,2022-Dec,2023

Project:

Credit Risk Forecasting:

- **As a Python/PySpark developer** - (Python/Pyspark) responsible for building an end-to-end data ingestion pipeline having developed various customized stringent checks with ETL on financial data ingesting to different storage including delta lake table.
- **Containerizing the spark applications** - and deploying on Kubernetes cluster having utilized various spark cluster options.
- **Performance tuning & Benchmarking-** on legacy spark code having implemented spark partition strategy, right APIs, cluster configuration etc.

Technology used - Python 3.9, PySpark 3.5.1, Google bigquery, Postgresql, AWS S3, NAS, delta lake,Docker, Kubernteets.

Data Engineer
Arup Consulting

Aug 2019 - 1st July 2022
London, UK

Projects:

- **Lead Data Engineer – GSA data pipeline:**

ETL project, creating an on-demand query support engine to be used by structural engineers.

- Lead Data Engineer in a cross-functional team including, civil engineers, data scientist
- Allowed engineers to visualize structural data and perform analysis (comparison and permutation) to quickly compare building design data on carbon calculation etc
- Used AWS Glue catalog and ETL (PySpark), Athena, Simba+athena driver to connect power BI for visualization.

- **Lead Data Engineer - Offshore Wind Foundation (OWF) Phase-I:**

Migrated a scientific wind power calculation software of offshore wind farm from on-prem based application to AWS cloud involving tools like Airflow, EMR(PySpark), Glue, Athena

- Accommodated more input data (scaling to terabytes)
- Improved time efficiency of energy calculation pipeline execution from 20 hours to 5 hours, improving energy production accuracy and frequency of production readings

- **Data Engineer - Offshore Wind Foundation (OWF) Phase-II:**

- As a team, achieved serverless containerization of existing applications on AWS along with orchestration through airflow cluster on Qubole (a third party data lake solution), resulting in improved deployability and better orchestration
- Created CI/CD pipeline on Azure DevOps, PowerBi dashboarding, AWS Athena tables

- **Lead Big Data Engineer - Arup Data Platform (ADP):**

Worked as a lead big data engineer/consultant in a cross-functional ADP team. Big data is one of the leading roles contributing to multiple consulting projects.

- Responsible for maintaining Qubole environment (spin up, executing, clean up) along with coordinating with Qubole tech support team, and platform management
- Involved in consulting on various client-facing big data projects
- Helped different internal and external team deliver their work

Technologies Used: PySpark, Python, SQL, Terraform, Azure DevOps, Azure storage, Azure VMs, AWS ECS, AWS managed Airflow, EMR, Glue.

- **Data Engineer - Insights from Projects (IFP):**

Worked with data scientists, data engineers, product managers and more working on carbon assessment and benchmarking.

- Solely responsible for big data engineering
- Creating a new tool to improve the quality of data analysis on Arup designed buildings, informing on post-code specific average carbon emissions and uncovering which buildings have the lowest carbon footprint etc
- Saved global Arup wide by designing a SQL query to quickly find carbon consumption as an alternative to coordinating by emails

Technologies Used: EMR, Glue, Athena, PySpark, Python, SQL, AWS lambda

Cloud Engineer/ Data Engineer
Cloudwick Technologies

Feb 2016 - July 2019
London, UK

Projects:

- **Cloud Engineer (Office for National Statistics, UK)**
Worked in a team dedicated to creating a platform allowing for research to be securely performed on the cloud.
 - Used services like Computing (EC2, Workspaces), storage (RDS, S3, EBS, DynamoDB), user management (Active Directory, IAM), Serverless (Lambda, API Gateway), Cloudberry, Network(VPC, Security Groups, NACLs), Cloud formation for single-click deployment etc.
- **Data Engineer (Smart Metering System, UK)**
Built Solutions to replace traditional reporting systems with on-prem databases.
 - Involving Homogenous and Heterogenous database migration using AWS DataBase Migration System and SCT, AWS Lambda for event-based triggering to do transformation (in Python) and for triggering AWS Glue jobs to perform ETL and metadata management. Plus, Athena for aggregation and powerBi for visualization.
- **Data Engineer (Nordea Bank, Europe)**
 - Created a solution from scratch for customized visualization on Hadoop Performance Testing with historical and live workflow monitoring
 - Created a Data Lake on HDFS for all performance metrics data. Used python (OOPs approach), Kafka, Oozie, Yarn Application, different REST APIs, JVM stats, Shell scripting, Hive, HDFS
- **UK Accident data (Cloudwick Internal POC)**
Created a Data Lake on HDFS for accident data. Used geocoding in python for location names.
 - Used Pyspark for analyzing the whole dataset. Tuning was performed for efficiency comparison performed between Hive on Tez and Hive on Spark.
- **POC for London Council of Waltham Forest (Cloudwick Internal POC)**
Developed POC to make pricing predictions on housing data considering two approaches:
 - 1: Designed and developed Machine Learning Model using AWS Machine Learning tool
 - 2: Designed and developed a Machine learning model for clustering (K-means) on spark and made predictions on that model.

Approach 1 can be used for making predictions for the house pricing for a given area.
Approach 2 was useful for finding the latent pattern among house prices.

Research Associate (Computer Science Department)
University of Leicester

July 2015 - Feb 2016
Leicester, UK

Research assistant with Prof (Dr.) Ruzanna Chitchyan on Social Sustainability Project.

- Created data structure and performed qualitative data analysis on Airbnb and Uber
- Responsible for data (textual) parsing with Webmatrix. Written program for NLP in python to handle tagged text data

Qualifications:

MSc in Computer Science

2014 - 2016

University of Leicester, UK

Bachelor of Engineering in Computer Science

2008 - 2011

The Institution Of Engineers, India

Diploma in Computer Science

2005 - 2008

West Bengal State Council of Technical Education (WBSCTE), India